ZACH Sweeney EXAM 3, 12:13

$\boxed{82}$

① total volume change

Given X_{th}, $\dot{F}$, T_p, T_{ret}, ΔP_0

$$\epsilon_{tot} = \epsilon_{TE} + \epsilon_D + \epsilon_{SFP} + \epsilon_{GFP} + \cancel{\epsilon_C}$$

9/10

$$\epsilon_{TE} = \alpha \Delta T = 10\cdot10^{-6}\cdot(1500-300) = \underline{0.012}$$

$$\epsilon_D = \Delta P_0 \exp\left(\frac{\beta \ln(0.01)}{C_D B_D} - 1\right) \quad \beta = \frac{\dot{F}t}{N_U} \to \text{Assume } N_U \approx 2.44\cdot10^{22}\ \frac{U}{cc}$$

$$\beta = \frac{8\cdot10^{13}\cdot(300\text{ days}\cdot24\cdot3600)}{2.44\cdot10^{22}} = 0.085 > B_D$$

$\frac{5 \cdot WD}{950} = 0.005$

$$\therefore \epsilon_D = \Delta P_0 = \underline{0.015}$$

$$\epsilon_{SFP} = 5.577\cdot10^{-2}\rho\beta = 5.577\cdot10^{-2}\cdot10.97\cdot(0.085)$$

$$\epsilon_{SFP} = \underline{0.052}$$

$$\epsilon_{GFP} = 1.96\cdot10^{-28}\cdot\rho\beta\,(2800-T)^{11.73}\cdot e^{-0.0162(2800-T)}\ e^{-17.8\rho\beta}$$

Plugging in... $T=1500$, $\rho = 10.97$, $\beta = 0.085$

$$\epsilon_{GFP} = \underline{1.050\cdot10^{-3}}$$

~ close...

$$\therefore \epsilon_{tot} = 0.012 + 0.015 + 0.052 + 1.05\cdot10^{-3}$$

~~$\epsilon_{tot} = 0.050$~~

$\boxed{\epsilon_{tot} \quad 0.08005}$ OR 8% swell

densification is negative!

seems a bit high, but could be end of life

② Find ε_{cr} 2

$$\dot{\varepsilon}_{ss} = A_0\left(\frac{\sigma}{G}\right)^n \exp\left(\frac{-Q}{kT}\right)$$

$$\dot{\varepsilon}_{ss} = 3.14\cdot10^{24}\left(\frac{60\text{ MPa}}{\underbrace{4.2519\cdot10^{10} - 2.2185\cdot10^{7}\cdot600K}_{29.2\cdot10^{3}\text{ MPa}}}\right)^{n \to 5} \exp\left(\frac{-2.7\cdot10^{5}}{8.314\cdot600}\right)$$

6/8

$$\underline{\dot{\varepsilon}_{ss} = 3.58\cdot10^{-13}}$$

$$\dot{\varepsilon}_{IR} = C_0\,\Phi^{c_1}\,\sigma_m^{c_2} = 1.654\cdot10^{-34}\cdot(\text{LHR}\cdot ~~\ldots~~)^{c_1}\cdot(60\text{ MPa})^{c_2}$$

$$\varepsilon_{IR} = 1.654\cdot10^{-34}\cdot(250\cdot 3E11)^{0.85}\cdot(60\text{ MPa})^{1}$$

$$\dot{\varepsilon}_{IR} = 6.173\cdot10^{-5}$$ – way too high, calculator?

$$\therefore \dot{\varepsilon}_{tot} = \dot{\varepsilon}_{ss} + \dot{\varepsilon}_{IR} = 3.58\cdot10^{-13} + 6.173\cdot10^{-5}$$

$$\underline{\dot{\varepsilon}_{tot} = 6.173\cdot10^{-9}\ \tfrac{1}{s}}$$ – where did 5 change to 9?

$$\therefore \varepsilon_{cr} = \dot{\varepsilon}_{tot}\cdot t$$

→ should be 11, some math error...

$$\varepsilon_{cr} = 6.173\cdot10^{-9}(200\text{ days}\cdot24\cdot3600)$$

$$\boxed{\varepsilon_{cr} = 0.107}$$

seems high... it is

3 Find f 4/8

$t = 250$ days

Sample 1 $T=1400$ $F=2\cdot10^{3}$

$a = 5\,\mu m$

$D = D_1 = 7.6\cdot10^{-6}\exp\left(\frac{-3.03}{k_BT}\right)$

Find D, τ

$D = 7.6\cdot10^{-4}\exp\left(\frac{-3.03}{8.314\cdot1400}\right)$

$D_T = D_1 + \cancel{D_2} + \cancel{D_3}$

- we have irradiation!

$D = 7.6\cdot10^{-16}$

$\tau_1 = \frac{Dt}{a^2} = \frac{7.6\cdot10^{-16}\cdot(250\cdot24\cdot3600)}{(0.5\cdot10^{-3})^2}$

$\tau_1 = 0.0657 < \pi^{-2} = 0.101$

$\tau_2 = \frac{7.6\cdot10^{-16}\cdot250\cdot24\cdot3600}{(\underbrace{40}_{\text{all same besides } a}\cdot10^{-3})^2} = 1.026\cdot10^{-5}$ - unit mistake 4×10^{-3}

$\therefore$ For Sample 1, $\tau > \pi^{-2}$ → this is $\tau < \pi^{-2}$

$\therefore f = 4\sqrt{\frac{Dt}{\pi a^2}} - \frac{3}{2}\left(\frac{Dt}{a^2}\right)$ - equations mixed

$\boxed{f_1 = 0.046}$ $\therefore$ $\boxed{4.6\%\text{ released Sample 1}}$

For Sample 2, $\tau < \pi^{-2}$ $\therefore f = 1 - \frac{0.662}{\frac{Dt}{a^2}}\left(1 - 0.92e^{\left(\frac{\pi^2 Dt}{a^2}\right)}\right)$

plugging in same as above, but $a = 40\,\mu m$,

$\boxed{f_2 = \cancel{0.0} \; 0.175}$ $\therefore$ $\boxed{17.5\%\text{ released in Sample 2}}$

(4.) U^{4+} dominates. If O/M ↑ above 2.0, U oxidizes to compensate. This is due to fisson product accumulating and altering O potential.

4/4

(5) Lower O/M in center, higher as you move out to rim of pellet.

- lower at pellet rim

Mo acts as a buffer. At lower T's, it consumes O but is metallic at higher temps. Behaves differently across pellet depending on temperature.

3/5

- oxygen potential, not T

(6) Thermal Conductivity
Melting Point
Creep Rate

3/3

(7) Soluble Oxides → causes swelling, ↓ K
~~W~~ Soluble Oxides → causes swelling
Metals → ~~[illegible]~~ slightly increases K
Volatiles → swelling (↓ K), can cause cladding corrosion
Noble Gases → can cause swelling & raise gap pressure, both lower K

10/10

ANS 5

(8) 1 → intragranular diffusion
↳ driven by concentration gradients, gas diffuse thru lattice to grain boundaries

8/8

2. → bubble Nucleation/growth
↳ at grain boundary, bubbles grow as more atoms arrive and interconnect

3 → transport to free surfaces
↳ once bubbles grow, some path is created (i.e. a gap) and the gas escapes

(9) A time-dependent deformation due to prolonged/constant stress. ~~[illegible]~~ A form of plastic deformation which occurs below σ_Y.

5/5

1 type → irradiation-induced creep from n flux, n's create point defects thru collisions. there is some absorption which causes permanent deformation.

(10) HBS forms at rim of pellet, where there is large burnup due to U-238 n-capture.

8/8

Porosity increases due to defects at high burnup, but polygonization cleans defects to create new grain boundaries

Slight increase to thermal conductivity in region as a result.

~~[illegible]~~ Fission gas gets retained in structure to an extent.

(11) Zr's Anisotropic structure causes vacancies to migrate to different planes, cause expansion to basal planes.

4/6 - not enough

Results in elongation ~~in~~ one direction, shrinking in another. At macroscopic level, Axial elongation & Radial shrink.

(12) creep closes pellet gap over time, then pellet expansion loads the cladding. Allows for some stress relief in fuel since def. is slow.

5/8 - needed more

(13)

Susceptible material ~~Corrosive environment~~	Corrosive environment ~~Susceptible material~~	enough stress	time
hardening, brittleness reducing ductility, fracture toughness	caused by volatile fission products, breach oxide layer	pellet expansion loads cladding, causing stress concentration	time needed for corrosive effects to breach oxide layer

9/12

- needed more everywhere

(14) Iodine meets cladding to form ZrI_4 which breaches ~~breaks~~ the oxide layer ~~[illegible]~~

4/5

Iodine content controls crack initiation & rate, ∴ at higher B, cladding is more susceptible - wanted more